

Introduction to Knot Theory

(6) Knots and Graph Theory

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Knot Diagram

A knot diagram is a generic projection with crossing data.
Consider each arc as an edge, and crossing as a vertex.

Definition

A **medial graph**(or Knot shadow) is an underlying graph of a knot diagram by ignoring overstrand and understrand.

- $M(D)$ is a 4-valent planar graph(Eulerian).
- A dual graph $M(D)^*$ is bipartite.
- $M(D)$ loses information of crossings.

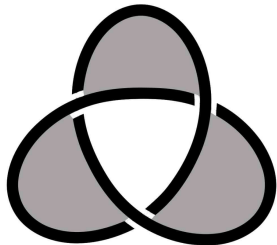
Checkerboard Coloring

Definition

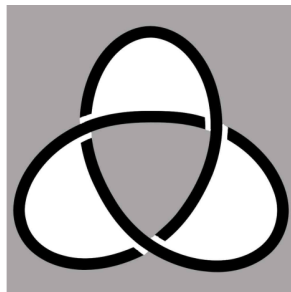
For a given diagram D , consider a medial graph of $M(D)$.

A **checkerboard coloring** (or checkerboard shading) is a way of coloring a face of $M(D)$ (or a vertex of dual graph of $M(D)$) with the unbounded face colored white.

We call dual checkerboard coloring when the coloring is inversed.



(a) Checkerboard



(b) Dual Checkerboard

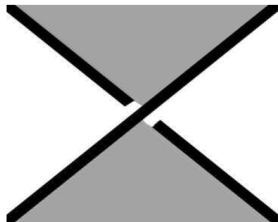
Tait Graph

Definition

For a given diagram D , consider a checkerboard coloring of D .

A **Tait graph** (from checkerboard coloring) $G(D)$ is a signed graph with vertex set of black colored faces from checkerboard coloring, and edge set of crossings.

A **signed Tait graph** $(G(D), \zeta)$ is a Tait graph with sign ζ of each edge given to be $+1$ if understrand is left to the surface and -1 if overstrand is left to the surface.



(a) $\zeta(c) = -1$



(b) $\zeta(c) = +1$

State–Graph correspondence

We can construct a bijection between each state and a subgraph of a (signed) Tait graph.

Proposition

For any state s of a given diagram D , let A_s be set of edges connecting black region in s . There is a bijection between state s and spanning subgraph of a Tait graph $G(D)$, given by $s \mapsto G_s := G(D)[A_s]$, where G_s is a spanning subgraph of $G(D)$ with edge set A_s .

$$\{s\} \leftrightarrow \{G_s := G(D)[A_s]\}$$

We can observe that for A-smoothing for negative sign crossing, and B-smoothing for positive sign crossing connects the black region.

Relations between States and Graphs

State-Graph Relations

$$|s| = k(G_s) + n(G_s),$$

where

$$n(G_s) = |A_s| - r(A_s)$$

is the nullity (cycle rank) of G_s .

Equivalently,

$$|s| = |A_s| + |V| - 2r(A_s) = |A_s| - |V| + 2k(G_s).$$

For a alternating knot, we can find a coloring (checkerboard or dual checkerboard) that gives every crossing a same sign $\zeta(e) = -1$. It makes A_s to be set of all A-smoothing edges.

Alternating Case

$$a(s) - b(s) = 2|A_s| - |E|$$

Tutte Polynomial

Definition

$$T(G, x, y) = \sum_{A_s \subseteq E} (x - 1)^{r(E) - r(A_s)} (y - 1)^{|A_s| - r(A_s)}$$

The Tutte polynomial is also the universal invariant encoding all subset expansions.

Uniqueness of Tutte polynomial

Theorem (Brylawski, 1972)

There is a unique function T from the class of graphs $\mathcal{G} = \{G = (V, E)\}$ into the polynomial ring $\mathbb{Z}[x, y]$ with

- ① $T((\{a, b\}, \{a \sim b\}), x, y) = x$, $T((\{a\}, \{a \sim a\}), x, y) = y$, and $T((V, \emptyset), x, y) = 1$.
- ② For any edge e is neither a isthmus of G nor a self-loop, then
$$T(G, x, y) = T(G - e, x, y) + T(G/e, x, y)$$
- ③ For any edge e is either a isthmus of G or a self-loop, then
$$T(G, x, y) = T(G[\{e\}], x, y) \cdot T(G/e, x, y)$$

The duality of A/B-smoothing, checkerboard coloring, and Tait graph is well preserved as
$$T(G, x, y) = T(G^*, y, x).$$

Skein Relations and Deletion–Contraction

Recursive structure of the Tutte polynomial mirrors the skein relation of the Kauffman bracket.

For a graph G ,

$$T(G, x, y) = T(G - e, x, y) + T(G/e, x, y)$$

whenever e is neither a bridge nor a loop.

For the Kauffman bracket,

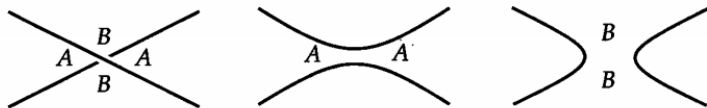


Figure: For each state K, K_A, K_B
 $\langle K \rangle = A \langle K_A \rangle + A^{-1} \langle K_B \rangle$

Deleting or contracting an edge in the Tait graph corresponds to choosing a smoothing.

Spanning tree expansion

The computation of Tutte polynomial doesn't depend on the ordering of edge.

We can fix the ordering on the edges of $G = (V, E)$, $E = \{e_0, \dots, e_m\}$ as $e_i < e_j$ for $i < j$.

Let $F = (V, E_F)$ be a spanning forest of G . For $e \in E_F$,

$$Cut_F(e) = \{f \in E - E_F \mid (V, E_F - \{e\} \cup \{f\} \text{ is a spanning forest of } G\}.$$

We say that $e \in E_F$ is internally active for F if $e < \min Cut_F(e)$.

For $e \in E - E_F$, $Cycle_F(e)$ is a unique cycle in $E_F \cup \{e\}$.

We say that $e \in E - E_F$ is externally active for F if $e = \min Cycle_F(e)$.

An (i, j) -forest F for G is a spanning forest with i edges internally active and j edges externally active.

Another version of Tutte polynomial

Theorem (Tutte, 1954)

Let G be a graph with ordering on its edges. Let $t_{i,j}(G)$ be number of (i,j) -forest in G , then

$$T(G, x, y) = \sum_{i,j} t_{i,j}(G) x^i y^j$$

By independence of the ordering

$$T(G, x, y) = \sum_{F : \text{spanning forest of } G} \left(\prod_{e \text{ internally active}} x \cdot \prod_{e \text{ externally active}} y \right)$$

$$T(G, x, y) = \sum_{F : \text{spanning forest of } G} x^{i(F)} y^{e(F)}$$

with $i(F)$ number of internally active edges of F , $e(F)$ number of externally active edges of F .

Another version of Tutte polynomial

Theorem (Kauffman 1989)

Let U be a Jordan curve of a diagram D . For each Jordan Euler trail T on U , it corresponds to maximal tree in planar graph $P(D)$. For a tait graph G , Tutte's theorem gives

$$T(G, x, y) = \sum_{T \text{ trails of } G} x^{i(T)} y^{e(T)}$$

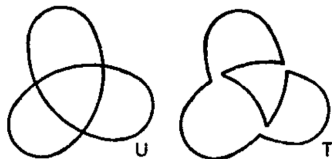


Figure: Jordan Euler trail T on a Jordan curve U

From the Kauffman Bracket to the Tutte Polynomial

Take $x = -A^{-4}$ and $y = -A^4$, then

$$\begin{aligned}T(G, x, y) &= \sum_{A_s \subseteq E} (x - 1)^{r(E) - r(A_s)} (y - 1)^{|A_s| - r(A_s)} \\&= \sum_{A_s \subseteq E} (-A^{-4} - 1)^{r(E) - r(A_s)} (-A^4 - 1)^{|A_s| - r(A_s)} \\&= \sum_{A_s \subseteq E} (A^{-2})^{r(E) - r(A_s)} (-A^{-2} - A^2)^{r(E) - r(A_s)} (A^2)^{|A_s| - r(A_s)} (-A^2 - A^{-2})^{|A_s| - r(A_s)} \\&= \sum_{A_s \subseteq E} (A^2)^{|A_s| - r(A_s) - r(E) + r(A_s)} \delta^{r(E) - r(A_s) + |A_s| - r(A_s)} \\&= \sum_{A_s \subseteq E} (A^2)^{|A_s| - |V| + 1} \delta^{|A_s| + |V| - 2r(A_s) - 1} \\&= \sum_{A_s \subseteq E} (A)^{2|A_s| - 2|V| + 2} \delta^{|s| - 1}\end{aligned}$$

From the Kauffman Bracket to the Tutte Polynomial

Remark (Kauffman bracket)

$$\langle D \rangle = \sum_s A^{a(s)-b(s)} \delta^{|s|-1}, \quad \delta = -A^2 - A^{-2}.$$

For alternating diagram,

$$\begin{aligned} \langle D \rangle &= \sum_s A^{2|A_s|-|E|} \delta^{|s|-1} \\ &= A^{2|V|-|E|-2} T(G, -A^{-4}, -A^4) \end{aligned}$$

Thistlethwaite Theorem

Theorem (Thistlethwaite, 1987)

Let D be a reduced alternating diagram with Tait graph G .

Then

$$V_L(t) = (-A^3)^{-w(D)} \langle D \rangle = (-t^{3/4})^{w(D)} t^{\frac{|E|-2|V|+2}{4}} T(G, -t, -t^{-1})$$

Therefore, for alternating links,

$$\text{Jones polynomial} \iff \text{Tutte polynomial}$$

Ribbon Graphs

Definition

A **ribbon graph** $\mathbb{G} = (V, E)$ is a surface with boundary, represented as the union of two sets of discs V and E with following conditions

- V and E intersect in disjoint union of line segments.
- For every intersection, there is a unique element of $v \in V$ and $e \in E$ s.t. $v \cap e$ be such line segment.
- For every $e \in E$, $V \cap e$ is disjoint union of exactly two line segments.

A ribbon graph consists of vertex discs, and edge bands attached to it.

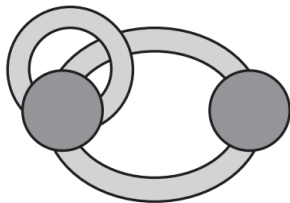
Two ribbon graphs are equivalent if there is a homeomorphism between two (orientation preserving if graph is orientable).

Arrow Presentation

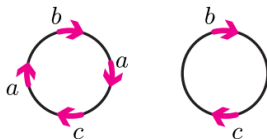
Definition

A **arrow presentation** of $\mathbb{G}$ consists of a set of circles and a set of labels. For each label there are exactly two arrows lying along one or more of the circles. All arrows are disjoint. The set of labels is called edges set.

The circles of arrow presentation is boundary of V and arrow(labeled arc) is intersection of E .



(a) Ribbon graph



(b) Arrow presentation

Ribbon Graph from a State

Given a state s of a diagram D and A_s be corresponding edge set connecting checkerboard shaded region. Similar to generating Tait graph, we can generate a ribbon graph $\mathbb{G}_s$. Choose checkerboarded shaded region as disjoint discs and attach a (untwisted) band in boundary for edges in A_s .

The ribbon graph forms an orientable surface with boundary components corresponding to state circles.

$$bc(\mathbb{G}_s) = |s|$$

Similarly, we can say

$$bc(\mathbb{G}_s) = |A_s| - |V| + 2k(\mathbb{G}_s).$$

Parameters of an Arrow Presentation

Let $\mathbb{G}$ be an arrow presentation with edge set E , and let $A \subseteq E$.

Definition

The ribbon graph has the following parameter agreed with one from the Tait graph.

$$e(A) := |A|,$$

$$v(A) := \text{number of circles in } \mathbb{G}(E \setminus A),$$

$$k(A) := \text{number of connected components of } \mathbb{G}(E \setminus A),$$

$$r(A) := v(A) - k(A).$$

Parameters of an Arrow Presentation

Definition

The ribbon graph has the some parameter originated from the 2-manifold structure.

$b(A) :=$ number of boundary components of $\mathbb{G}/A$,

$\gamma(A) := 2k(A) - v(A) + e(A) - b(A)$.

The parameter $\gamma(A)$ is the Euler genus of the ribbon graph associated to $\mathbb{G}(E \setminus A)$.

Let Σ_A be surface corresponding to $\mathbb{G}(E \setminus A)$. The Euler characteristic of Σ_A as cell complex and as a compact surface with $k(A)$ connected components are same

$$v(A) - e(A) + b(A) = \chi(\Sigma_A) = 2k(A) - \gamma(A)$$

Since we've constructed an orientable surface, $\gamma(A) = 2g$ for the genus g surface.

We call a ribbon graph is plane if $\gamma(E) = 0$, and the topology agrees with ordinary planar graph theory. We can define a polynomial that coincides with Tutte polynomial in planar case.

Ribbon Graph Polynomial

Recall the Tutte polynomial of a graph $G = (V, E)$:

$$T(G, x, y) = \sum_{A_s \subseteq E} (x - 1)^{r(E) - r(A_s)} (y - 1)^{|A_s| - r(A_s)}.$$

For ribbon graphs, the rank is modified to include topological data.

Define

$$\rho(A_s) = r(A_s) + \frac{1}{2}\gamma(A_s).$$

Definition

The ribbon graph polynomial of $\mathbb{G}$ is

$$R(\mathbb{G}, x, y) = \sum_{A_s \subseteq E} (x - 1)^{\rho(E) - \rho(A_s)} (y - 1)^{|A_s| - \rho(A_s)}.$$

Ribbon Graph Polynomial

- The ribbon graph polynomial extends the classical Tutte polynomial by incorporating the topology of the embedding surface.
- When $\mathbb{G}$ is planar,

$$\gamma(A_s) = 0 \quad \text{for all } A_s \subseteq E,$$

hence

$$R(\mathbb{G}, x, y) = T(G, x, y).$$

Seifert Algorithm

Recall (Seifert Algorithm)

Given an oriented knot diagram D :

- ① Smooth every crossing according to the orientation.
- ② Obtain a collection of Seifert circles.
- ③ Attach twisted bands at the crossings.

The resulting orientable surface S_D satisfies

$$\partial S_D = K.$$

Ribbon Graph Interpretation of Seifert Surfaces

The Seifert surface can be viewed combinatorially as a ribbon graph.

- Seifert discs correspond to vertices.
- Bands correspond to edges.
- Boundary components correspond to link components.

The topology of the surface is encoded by

$$\gamma(\mathbb{G}) = 2k(\mathbb{G}) - v(\mathbb{G}) + e(\mathbb{G}) - bc(\mathbb{G}).$$

The formula agrees with the genus formula of Seifert surface, since the variables are mapped as $k(\mathbb{G}) = 1$, $v(\mathbb{G}) = s$, $e(\mathbb{G}) = c$, and $bc(\mathbb{G}) = |\{K\}| = 1$ gives genus of Seifert surface

$$g = \frac{c - s + 1}{2}.$$

Bollobás–Riordan Polynomial

Definition

For a ribbon graph $\mathbb{G}$, a **Bollobás–Riordan polynomial** is

$$B(\mathbb{G}, x, y, z) = \sum_{A_s \subseteq E} (x-1)^{r(E)-r(A_s)} y^{n(A_s)} z^{k(A_s)-bc(A_s)+n(A_s)}.$$

For planar ribbon graph $\mathbb{G}$,

$$B(\mathbb{G}, x, y, 1) = R(G, x, y+1) = T(G, x, y+1).$$

Thus ribbon graphs generalize the graph-theoretic interpretation of knot invariants to non-planar settings.

Signed Ribbon Graphs

For arbitrary knot diagrams, crossing signs are mixed, so the sign of each crossing must also be recorded in the polynomial

Definition

Let $\hat{\mathbb{G}} = (\mathbb{G}, \zeta)$ be a signed ribbon graph with sign function

$$\zeta : E(\mathbb{G}) \rightarrow \{+1, -1\}.$$

Signed Bollobás–Riordan Polynomial

Definition

A **signed Bollobás–Riordan polynomial** is

$$B_{\zeta}(\mathbb{G}, x, y, z) = \sum_{A_s \subseteq E} (x-1)^{r(E)-r(A_s)} y^{n(A_s)-s(A_s)} z^{k(A_s)-bc(A_s)+n(A_s)},$$

where

$$s(A_s) = \frac{1}{2}(e_-(A_s) - e_-(E \setminus A_s)),$$

and $e_-(A_s)$ denotes the number of negative edges in A_s .

Kauffman Bracket and Signed polynomial

Theorem (Chmutov 2009)

Let L be a checkerboard colorable link diagram and $\widehat{\mathbb{G}}$ the associated signed ribbon graph. Then

$$[L](A, B, d) = A^{n(\widehat{\mathbb{G}})} B^{r(\widehat{\mathbb{G}})} d^{k(\widehat{\mathbb{G}})-1} B_{\zeta} \left(\widehat{\mathbb{G}}, \frac{Ad}{B} + 1, \frac{Bd}{A}, \frac{1}{d} \right).$$

For the Kauffman bracket,

$$B = A^{-1}, \quad d = -A^2 - A^{-2}.$$

Hence:

Kauffman bracket $\iff$ signed Bollobás–Riordan polynomial
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Theorems

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Images

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